1️⃣ Descriptor 9 (Most Relevant)

> **Understand the relationship between the context of the problem and the representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution.**

How it is reflected:

- Students must understand:
  - The horizontal axis represents months (the time order is critical)
  - Different coloured bars within each month represent different bands
  - The comparison is between **No One's Darling** and **The Kicking Kangaroos**
- The phrase **"for the first time"** introduces a contextual, time-based condition:
  - Students must examine the data sequentially from **January to June**
  - And identify the **first** month in which
    No One's Darling > The Kicking Kangaroos

✅ **This is the core and indispensable capability assessed by the question.**

2️⃣ Descriptor 5

> **Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation.**

How it is reflected:

- Within a **single graphical representation**, students must:
  - Compare the heights of two sets of bars
  - Determine which value is greater
- This is a typical example of:
  - **Graphical information → comparative relationship → verbal conclusion (answer option)**

Although only one graph is used, the task clearly involves **comparison and evaluation of data**, rather than simple data reading.

3️⃣ Descriptor 1

> **Interpret a mathematical result back into the real-world context.**

How it is reflected:

- The final answer is not a number, but a **month**
- Interpreted in real-world terms as:

  > "This is the month in which No One's Darling's CD sales first exceeded those of The Kicking Kangaroos."

That is:
a mathematical comparison → a temporal event → a real-world statement.

4️⃣ Descriptor 7

> **Construct and communicate explanations and arguments in the context of the problem.**

How it is reflected (implicitly but clearly):

- A complete reasoning process would involve:
  - Comparing the two bands' sales month by month
  - Explaining why earlier months do not satisfy the condition
  - Identifying the month that satisfies the condition for the first time

For example:

> "Before March, No One's Darling sold fewer CDs than The Kicking Kangaroos, and in April it sold more for the first time, so the answer is April."

⚠️ Although an explicit explanation is not required in a multiple-choice format,
the **logical structure of the argument is essential**.

5️⃣ Descriptor 2

> **Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem.**

How it is reflected:

- Students must judge:
  - Whether there is indeed a month in which the condition is met
  - Or whether the correct choice should be **A. No month**
- This involves assessing whether a proposed conclusion is **consistent with the data shown**.

Thus, the task involves a **limited but genuine element of reasonableness checking**.

① 第 9 条（最核心）

> **Understand the relationship between the context of the problem and representation of the mathematical solution.**

具体体现：

- 学生必须理解：
  - 横轴是月份（时间顺序很关键）
  - 每个月中，不同颜色柱子代表不同乐队
  - 比较的是 **No One's Darling vs The Kicking Kangaroos**
- "for the **first time**" 是一个**情境性时间条件**
  - 要在图中从 **January → June** 依次检查
  - 找到**第一次**出现 "No One's Darling > The Kicking Kangaroos" 的月份

👉 若不能正确建立**情境—图表—时间顺序**之间的关系，无法作答。

✅ **这是该题最核心、不可替代的能力点**

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② 第 5 条

> **Interpret mathematical outcomes in a variety of formats … compare or evaluate two or more representations in relation to a situation**

具体体现：

- 学生需要在**同一图表中**：
  - 比较两组柱形数据的高度
  - 判断"谁大于谁"
- 属于典型的：
  - **图像 → 比较关系 → 语言判断（选项）**

虽然仍然只有一种图表形式，但已经涉及**数据之间的比较与评估**，而非简单读取。

③ 第 1 条

> **Interpret a mathematical result back into the real world context**

具体体现：

- 最终答案不是一个数，而是一个**月份**
- 该月份在现实语境中的解释是：

  > "这是 No One's Darling 的 CD 销量首次超过 The Kicking Kangaroos 的月份。"

即：

比较结果 → 时间事件 → 现实描述

④ 第 7 条

> **Construct and communicate explanations and arguments in the context of the problem**

具体体现（隐含但明显）：

- 一个完整的解题逻辑应为：
  - 逐月比较两支乐队的销量
  - 说明前几个月为何不满足条件
  - 指出在哪一个月首次满足

例如：

> “在 3 月之前 No One’s Darling 都低于 The Kicking Kangaroos，而在 4 月首次高于，因此答案是 April。”

⚠️ 在选择题中不要求写出，但**推理结构是必须的。**

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⑤ 第 2 条

> **Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem**

具体体现：

- 学生需要判断：
  - 是否存在“首次超过”的月份
  - 或是否应选择 **A. No month**
- 这实际上是在判断：
  - “得出某个月”这一结论是否**符合图表事实**

因此，该题**轻度涉及合理性判断**，但仍基于直接数据。